

Self-Learning: Db2 on Cloud, Backup & Recovery, Indexing & Warehouse DBMS Module 6

> **Definition:** Enterprise Db2 deployments utilize **Db2 on Cloud**, **Db2 Warehouse**, automated **Backup & Recovery utilities**, and **B+ Tree Indexing** for cloud scalability, data protection, and high-performance warehousing.

1. Detailed Technical Explanation

1. Db2 Backup & Recovery Utilities

IBM Db2 provides command-line utilities to perform full, incremental, and delta backups to protect against hardware failures and data corruption.

```
```bash
```

```
db2 BACKUP DATABASE STUDENTDB TO "C:\Db2Backups"
```

### 2. Online Database Backup (Allows active user transactions while backing up)

```
db2 BACKUP DATABASE STUDENTDB ONLINE TO "C:\Db2Backups" INCLUDE LOGS
```

### 3. Database Restore Operation

```
db2 RESTORE DATABASE STUDENTDB FROM "C:\Db2Backups" TAKEN AT 20260816120000
```

### 4. Rollforward Recovery (Replays transaction log records after restore)

```
db2 ROLLFORWARD DATABASE STUDENTDB TO END OF LOGS AND COMPLETE
```

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## 2. Db2 Indexing & Performance Tuning

Indexes in Db2 use **B+ Tree data structures** to accelerate lookup performance.

```
```sql
```

```
-- Create a Composite Index for fast searching on Last Name & First Name
```

```
-- Reorganize table data to reclaim fragmented space
```

```
-- Update optimizer statistics for accurate query planning
```

3. Db2 on Cloud & Db2 Warehouse on Cloud

1. IBM Db2 on Cloud:

- Fully managed Cloud Database-as-a-Service (DBaaS) hosted on IBM Cloud and AWS.

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2. IBM Db2 Warehouse on Cloud:

- High-performance columnar analytics warehouse powered by **BLU Acceleration**.

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4. Core Concepts & Memory Keywords

- **db2 BACKUP` / `RESTORE`:** Core command line utilities for disaster recovery.

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5. Must-Write Points for Exams

- **`RUNSTATS`** must be executed after major data loads so the Db2 query optimizer can choose optimal execution plans.

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6. Quick Recall Flow

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